WBC TUTOR LMS — WATER SECTOR SKILLS DEVELOPMENT PROGRAMME

Functions and Graphs

A Practical, Self-Paced Course for Reading, Sketching and Applying Real-World Graphs

Course Overview

Every pump curve, dosing chart, SCADA trend and reservoir-filling record is a graph. This self-paced course builds the skill of reading, sketching and applying functions and graphs from the ground up — covering the Cartesian plane, linear, quadratic and exponential functions (Algebraic functions)— using clear explanations, fully worked examples, and, throughout, real water-sector scenarios that show exactly why each graph matters on the job. No prior graphing experience is assumed.

Who Is This Course For?

- Mid-secondary scholars (ages 16–18) looking to revise and strengthen their functions and graphs before exams or further study.
- Water-sector technicians and operational staff who need to read and interpret pump curves, dosing charts, SCADA trends and other technical graphs as part of their daily work.
- Adult learners returning to study who want a practical, jargon-free refresher in graphing and functions.
- Learners who have completed (or are completing) the Introduction to Algebra course and want to build on those skills by applying algebra visually, as graphs.
- Anyone preparing for further technical or vocational training that assumes solid graph-reading fundamentals.

Course Purpose & Structure

This course takes learners on a carefully sequenced journey — from plotting points on the Cartesian plane and understanding what a function is, through to confidently sketching, interpreting and applying linear, quadratic and exponential graphs to real technical problems. It is delivered entirely online via the WBC Tutor LMS, so learners can study at their own pace, revisit any topic at any time, and track their own progress.

Every one of the 9 topics follows the same consistent, easy-to-follow structure:

- Learning Objectives — know exactly what you'll be able to do by the end of the topic.
- Definition & Explanation — plain-language teaching, with clearly labelled diagrams and figures.
- Key Characteristics — a consistent, six-point checklist (intercepts, zeros, turning points, asymptotes, domain and range, end behaviour) applied to every graph type.
- Worked Examples — step-by-step solutions, including realistic water-sector applications such as pump curves, reservoir filling and chlorine decay.
- Self-Study Practice — problems to attempt independently, with full worked solutions provided.
- Tips for Learners — practical advice to help you avoid the most common mistakes.
- Diagnostic Quiz — a comprehensive quiz covering every topic taught, so you can check your understanding before moving on.

A companion Summary sheet is also provided, comparing linear, quadratic and exponential functions side by side on a single page — ideal for ongoing use once the course is complete. Altogether, the course represents approximately 24 to 32 hours of self-paced study. The total time allocated for online access to complete the self-study course on the WBC LMS is two (2) months.

What You'll Learn — Topics & Objectives

The course progresses through 9 topics, building steadily from foundational graph-reading skills to real-world problem solving:

No.	Topic	What You'll Learn
1	**The Cartesian Plane**	Plot points using ordered pairs, identify quadrants, and read technical charts such as SCADA trends and pump performance curves.
2	**What Is a Function?**	Apply the vertical line test to distinguish functions from non-functions, and explain why an unambiguous relationship matters in water treatment control.
3	**Dependent and Independent Variables**	Identify the independent and dependent variable in a real relationship, and read and interpret data given as a table of values.
4	**Domain and Range**	Determine the domain and range of linear, quadratic and exponential functions, including real-world (contextual) restrictions.
5	**Function Notation**	Write equations in f(x) notation, evaluate a function at a given input, and discuss domain and range using function notation.
6	**Characteristics of a Function**	Identify intercepts, zeros, turning points, asymptotes, axis of symmetry and end behaviour using one consistent checklist.
7	**Linear Functions**	Sketch and interpret straight-line graphs, calculate gradients, determine equations from points or graphs, and apply linear models to flow and pressure scenarios.
8	**Quadratic Functions**	Sketch parabolas, find roots, vertex and axis of symmetry both algebraically and graphically, and apply quadratic models to pump and trajectory problems.
9	**Exponential Functions**	Recognise growth and decay, determine an equation from a graph, and apply exponential decay models to chlorine residual and similar water-quality scenarios.

Course Policies & Practical Information

1. Engagement with Facilitator

The learner / participant will be able to post questions and/or comments on the activities / tasks. The developer of this self-study course will regularly respond to the questions and comments. The developer may also arrange an online meeting to respond to the questions and/or comments. The date and time for such an online meeting, specific to a course, will be posted on the WBC LMS. All online meetings / forums, etc. may / will be recorded and posted on the WBC LMS for the benefit of all learners / participants registered for the course.

2. Flexibility / Accessibility of Course Material

The self-study courses are available 24/7 and the learner / participant can start at any time once registered!

3. Course Structure

The content for this self-study course is sub-divided based on specific topics. The accompanying text and associated worked examples will guide learners through various topics including tasks / activities to be studied / completed by the learner. The course concludes with Self-Study Exercises and a Diagnostic Quiz. Learners are expected to achieve 100% for the quiz.

4. Refunds

Cancelling and/or not completing the course does not qualify the learner / participant to be eligible for a refund. Refunds will only be considered if there are compelling reasons (for cancellation / termination) beyond the control of the learner / participant.

5. In Addition

- The Foundation Courses are not related to any qualification but may serve to support specific content of a related qualification.
- The Foundation Courses are not associated with specific NQF levels.
- This Foundation Courses are not accredited by the Engineering Council of South Africa (ECSA) and/or the South African Council for Natural Scientific Professions (SACNASP).

6. Course Fees

The introductory fee for the online **Functions and Graphs** self-study course **for employed staff is R 250**.

The **significantly reduced introductory fee** for the online **Functions and Graphs** self-study course **for registered (current registrations) full-time and part-time scholars** at any education institution is **R 50**. Scholars will be required to provide a copy of their annual registration form (current year) issued by the educational institution. Please submit your documentation to: students@waterbusinesscollege.co.za.

Ready to see mathematics come to life on a graph?

Enrol today through the WBC Tutor LMS and start learning at your own pace.

Please note: For this pilot phase only your full names & email address is required